

Macro-level bandwidth model for monolithic 3D memory

Terms

C, A, L	capacity, footprint/layer, layers
$\Delta T_{\max}$	peak ΔT (thermal governor)
N_{ind}	indep. peripheral sets
N_{sh}	arrays sharing sense amps
N_{tot}	$= N_{\text{ind}} \cdot N_{\text{sh}}$
$N_{\text{SA}}, N_{\text{dec}}$	# sense amps, # decoders
b_{acc}	bits per access (sense width)
$N_{\text{BL}}, N_{\text{WL}}$	bitlines, wordlines / array
t_{cyc}	steady-state cycle time
$t_{\text{dec}}, t_{\text{WL}}, t_{\text{BL}}$	decode (max stage latency), wordline, BL-develop
t_{SA}	sense (<i>incl. restore</i>)
t_{sw}	selector switching
$\text{Vol}_{\text{arr}}, \text{Vol}_{\text{SA}}$	array, sense-amp volume
$\text{Vol}_{\text{dec}}, \text{Vol}_{\text{sel}}$	decoder, selector volume

Cycle time

$$t_{\text{cyc}} = \max\left(t_{\text{dec}}, t_{\text{SA}} + t_{\text{sw}}, \frac{t_{\text{dec}} + t_{\text{WL}} + t_{\text{BL}} + t_{\text{SA}} + t_{\text{sw}}}{N_{\text{sh}}}\right)$$

Bandwidth

$$\text{BW} = \frac{b_{\text{acc}}}{t_{\text{cyc}}} N_{\text{ind}} \quad b_{\text{acc}} \leq N_{\text{BL}}$$

Counts, volume, scaling

$$N_{\text{tot}} = N_{\text{ind}} N_{\text{sh}}, \quad N_{\text{SA}} = N_{\text{ind}} b_{\text{acc}}, \quad N_{\text{dec}} = N_{\text{ind}}$$

$$\text{Vol} = \text{Vol}_{\text{arr}} N_{\text{tot}} + \text{Vol}_{\text{SA}} N_{\text{SA}} + \text{Vol}_{\text{dec}} N_{\text{dec}} + \text{Vol}_{\text{sel}} (N_{\text{sh}})$$

$$\text{Vol}_{\text{dec}} \propto N_{\text{WL}} N_{\text{BL}} \propto \text{Vol}_{\text{arr}}, \quad \text{Vol}_{\text{SA}} \propto \frac{1}{\text{margin}^2}$$

$$t_{\text{WL}} = k_{\text{WL}}^{\text{w}} N_{\text{BL}}^2 + k_{\text{WL}}^{\text{c}} N_{\text{BL}} \quad (\text{wire self-RC} + \text{cell load})$$

$$t_{\text{BL}} = k_{\text{BL}}^{\text{w}} N_{\text{WL}}^2 + k_{\text{BL}}^{\text{c}} N_{\text{WL}} \quad (k^{\text{w}} \text{ shared}, k^{\text{c}} \text{ per-tech})$$